THE TUBE LEMMA: AN OPEN NEIGHBOURHOOD OF A PRODUCT OF QUASI-COMPACT SUBSETS CONTAINS A PRODUCT OF OPEN NEIGHBOURHOODS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The product topology and the notion of a basic open neighbourhood of a point of the product, together with the quasi-compactness of the two factors. Set up in the author's sections on products and on quasi-compact spaces, before the counted statement. Supplies the basic product neighbourhoods used at the start of the proof and the finite subcovers extracted from the two quasi-compact sets.

Lemma 0.1 (Tube lemma)**.** *Let X and Y be topological spaces. Let $A \subset X$ and $B \subset Y$ be quasi-compact subsets. Let $A \times B \subset W \subset X \times Y$ with W open in $X \times Y$. Then there exists opens $A \subset U \subset X$ and $B \subset V \subset Y$ such that $U \times V \subset W$.*

Proof. For every $a \in A$ and $b \in B$ there exist opens $U_{(a,b)}$ of X and $V_{(a,b)}$ of Y such that $(a, b) \in U_{(a,b)} \times V_{(a,b)} \subset W$. Fix b and we see there exist a finite number $a_1, \ldots, a_n$ such that $A \subset U_{(a_1,b)} \cup \ldots \cup U_{(a_n,b)}$. Hence

$$A \times \{b\} \subset (U_{(a_1,b)} \cup \ldots \cup U_{(a_n,b)}) \times (V_{(a_1,b)} \cap \ldots \cap V_{(a_n,b)}) \subset W.$$

Thus for every $b \in B$ there exists opens $U_b \subset X$ and $V_b \subset Y$ such that $A \times \{b\} \subset U_b \times V_b \subset W$. As above there exist a finite number $b_1, \ldots, b_m$ such that $B \subset V_{b_1} \cup \ldots \cup V_{b_m}$. Then we win because $A \times B \subset (U_{b_1} \cap \ldots \cap U_{b_m}) \times (V_{b_1} \cup \ldots \cup V_{b_m})$. □

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